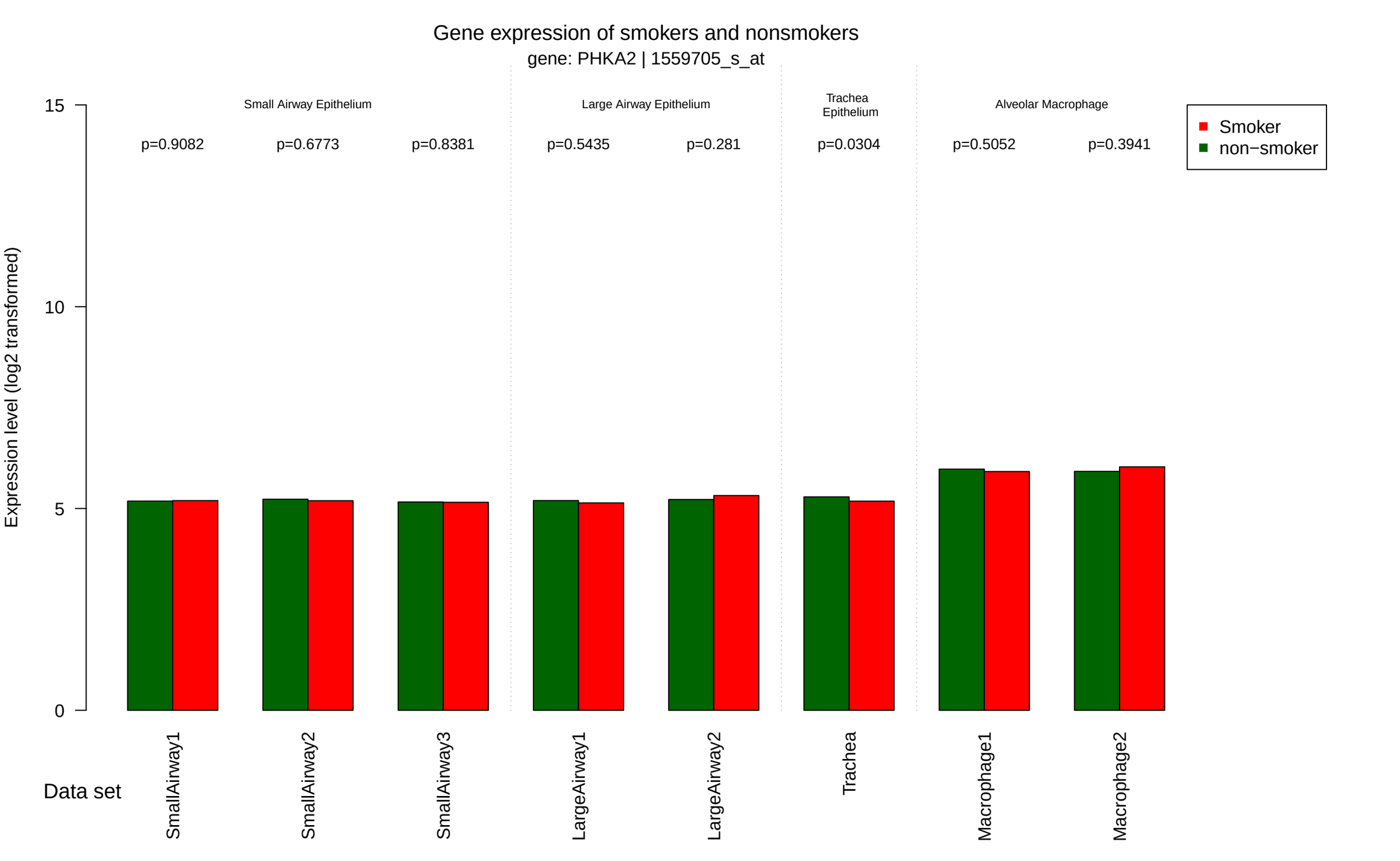
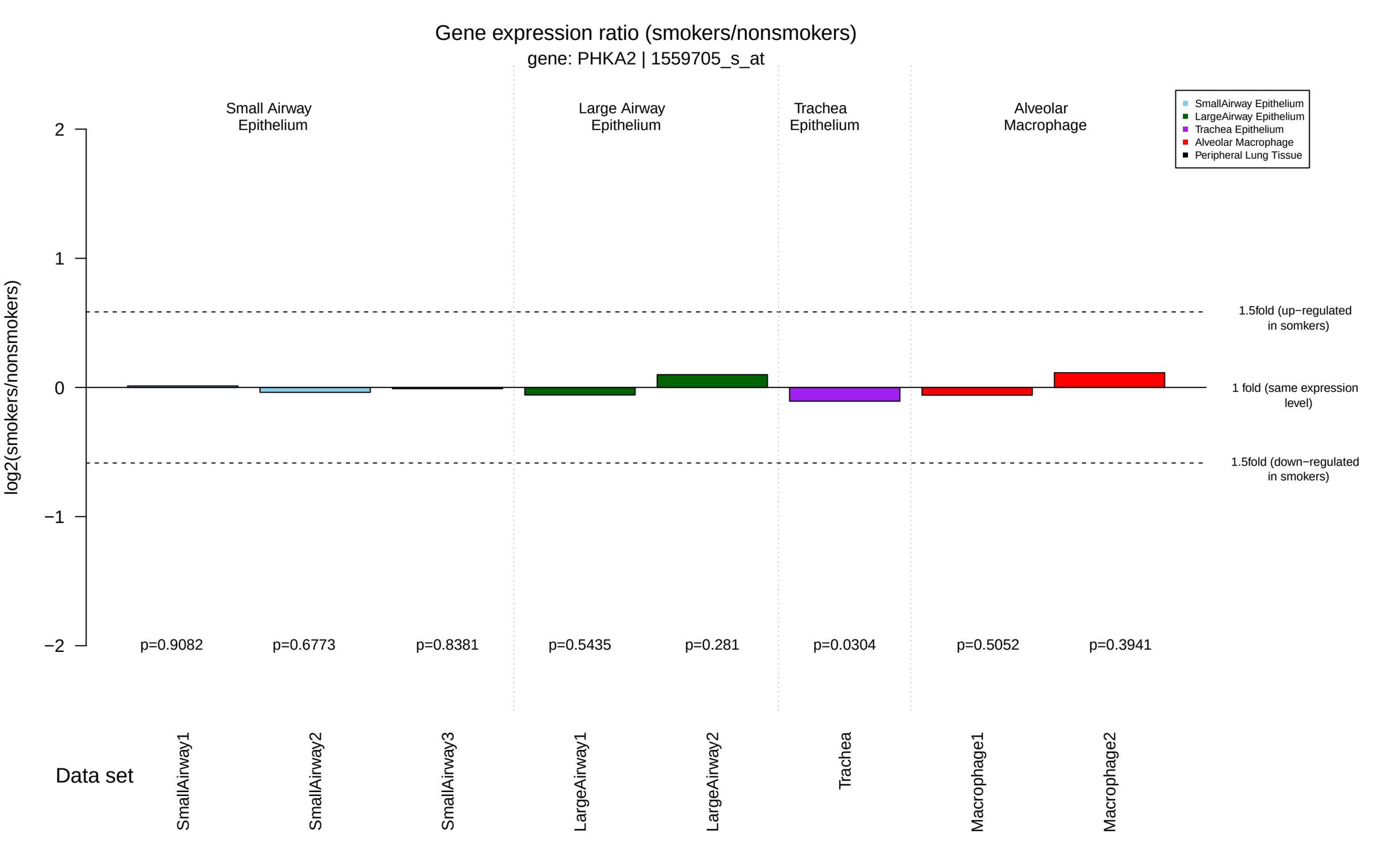
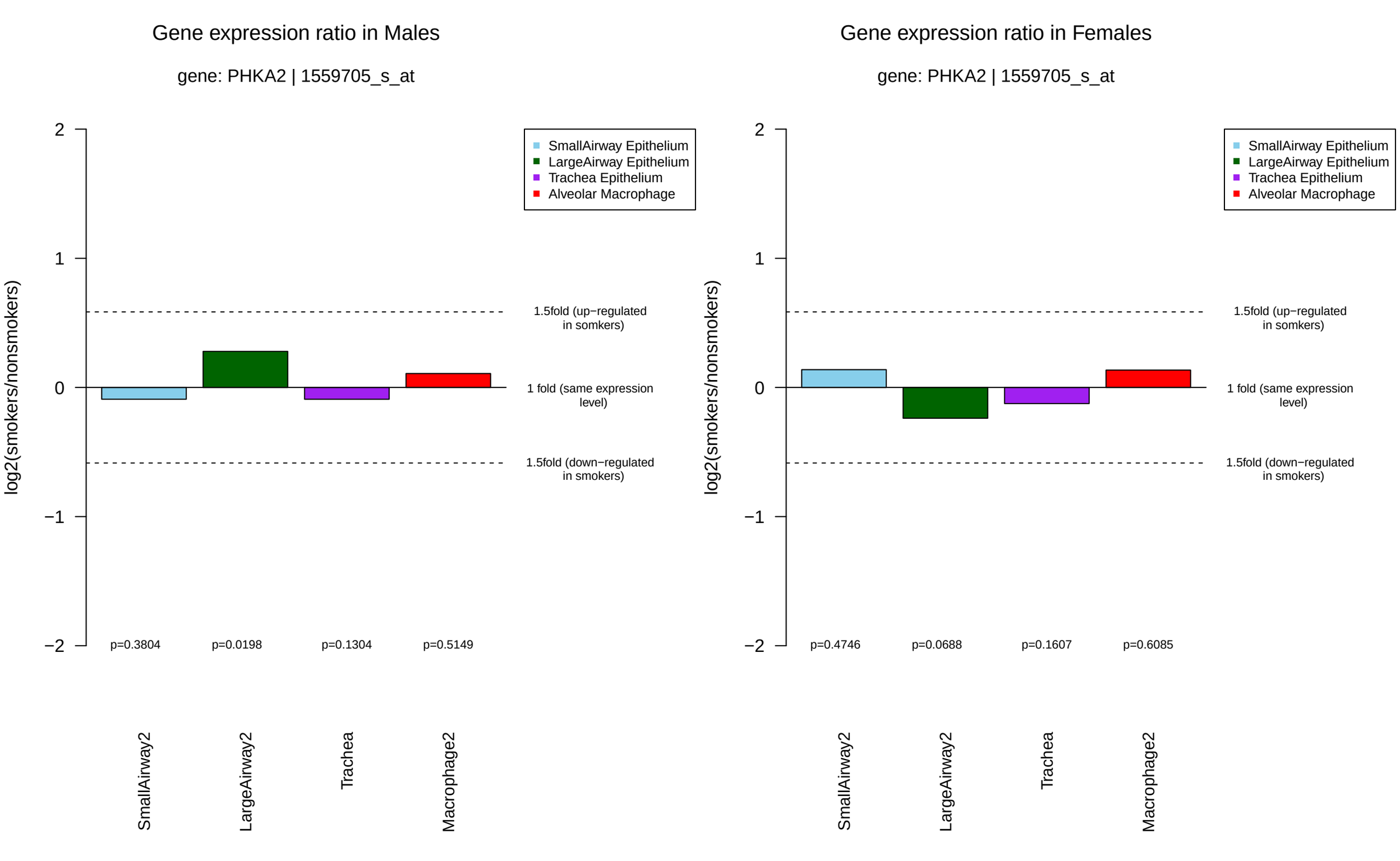
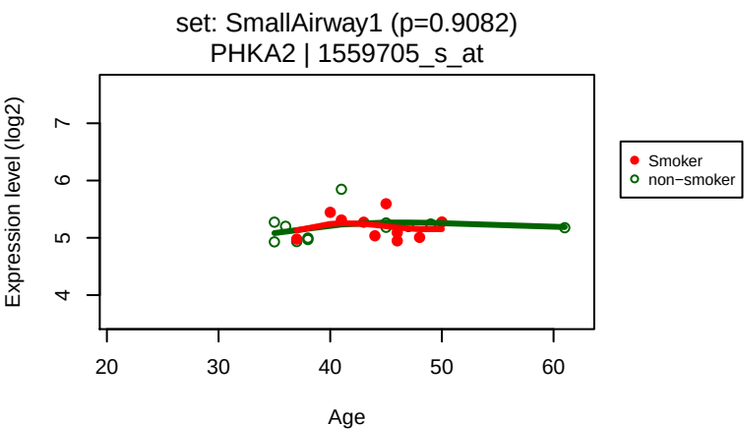
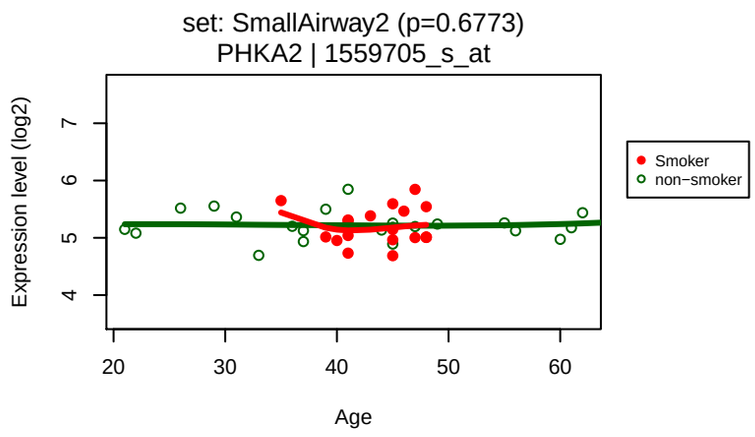
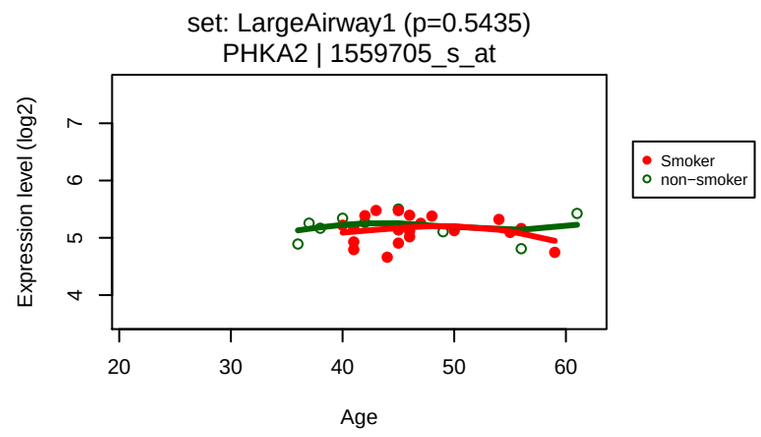

Gene expression of smokers and nonsmokers
gene: PHKA2 | 1559705_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9082
p=0.6773
p=0.8381
p=0.5435
p=0.281
p=0.0304
p=0.5052
p=0.3941
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

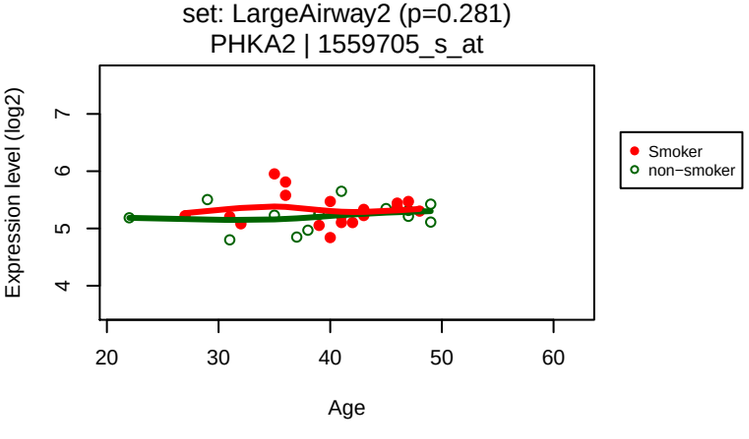

Gene expression ratio (smokers/nonsmokers)
gene: PHKA2 | 1559705_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9082
p=0.6773
p=0.8381
p=0.5435
p=0.281
p=0.0304
p=0.5052
p=0.3941
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

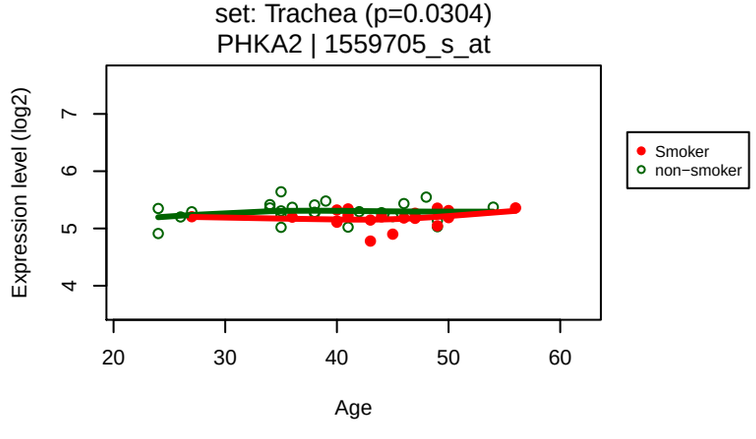

Gene expression ratio in Males
gene: PHKA2 | 1559705_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3804
p=0.0198
p=0.1304
p=0.5149
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PHKA2 | 1559705_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4746
p=0.0688
p=0.1607
p=0.6085
SmallAirway2
LargeAirway2
Trachea
Macrophage2

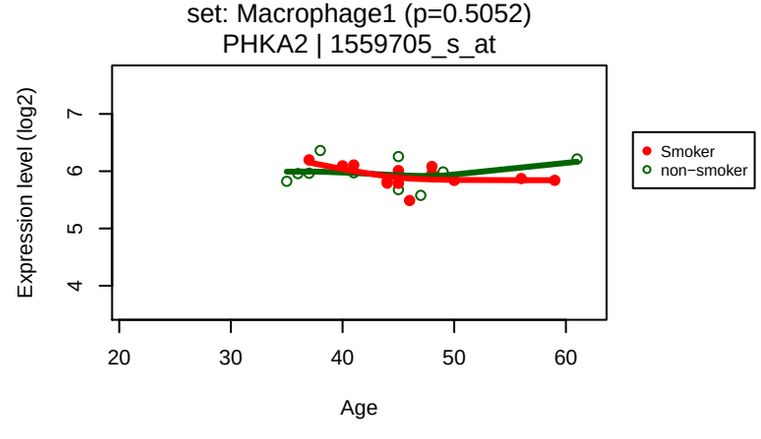

set: SmallAirway1 (p=0.9082)
PHKA2 | 1559705_s_at
Expression level (log2)
Age
Smoker
non−smoker

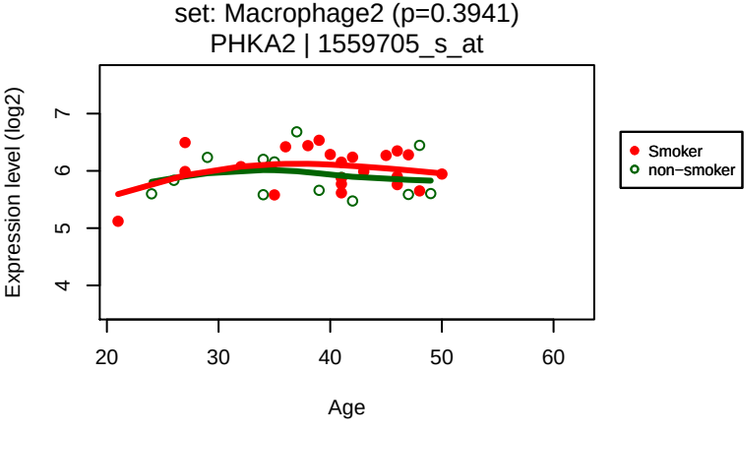

set: SmallAirway2 (p=0.6773)
PHKA2 | 1559705_s_at
Expression level (log2)
Age
Smoker
non−smoker

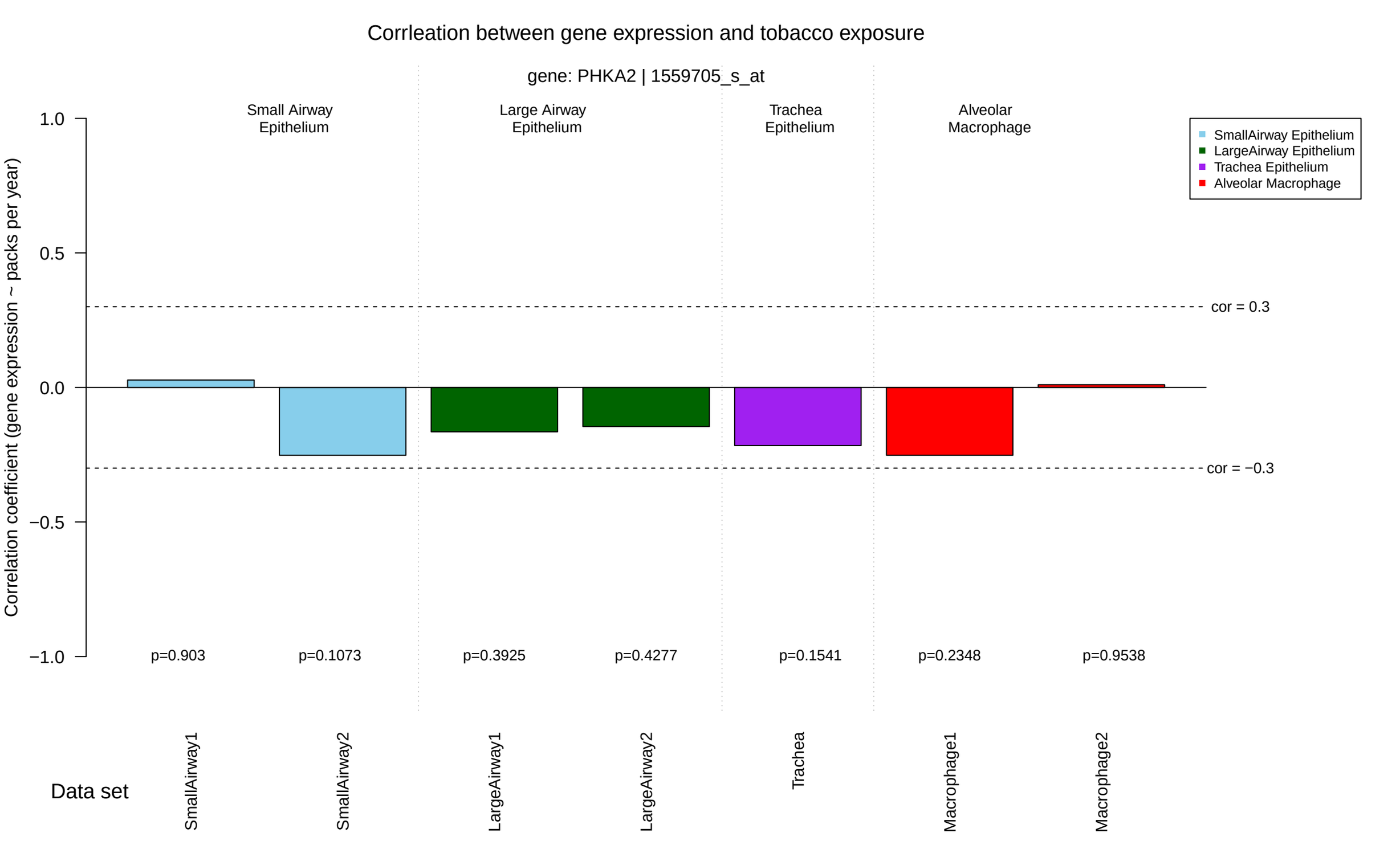

set: LargeAirway1 (p=0.5435)
PHKA2 | 1559705_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.281)
PHKA2 | 1559705_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0304)
PHKA2 | 1559705_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5052)
PHKA2 | 1559705_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3941)
PHKA2 | 1559705_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PHKA2 | 1559705_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.903
p=0.1073
p=0.3925
p=0.4277
p=0.1541
p=0.2348
p=0.9538
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2